

INFOSYS SPRINGBOARD
VIRTUAL INTERNSHIP 7.0
INTERNSHIP COMPLETION REPORT
PROJECT REPORT
GROUP – 1
BATCH 2

INTERN NAME: MOSURI SAHITHI GREESHMA

MENTOR NAME: MEGHA M

INTERNSHIP DURATION: 8 WEEKS

START DATE: 29 JUNE 2026

PROJECT TITLE: CYBER THREAT ANALYTICS

PLATFORM WITH INCIDENT VISUALIZATION SYSTEM

PROJECT TITLE :

Cyber Threat Analytics Platform with Incident Visualization System

Project Objective:

The main objectives of the Cyber Threat Analytics Platform with Incident Visualization System are:

- To collect and integrate incident and emergency response datasets for analysis.
- To clean, transform, and prepare the datasets using Power Query.
- To analyze incident categories, severity levels, and temporal patterns.
- To evaluate emergency response strategies and operational factors.
- To assess risk levels, threat exposure, and operational vulnerabilities.
- To identify resource allocation gaps and high-risk incidents.
- To develop interactive Power BI dashboards for effective visualization and analysis.
- To generate actionable insights and support data-driven operational decision-making.

PROJECT DESCRIPTION

Project Overview

The **Cyber Threat Analytics Platform with Incident Visualization System** is an analytics and visualization project developed to analyze incident, emergency response, and risk-related data using **Microsoft Power BI**.

The project uses two datasets, namely the **Indian Roads Dataset** and the **Integrated Emergency Response Dataset (IERAD)**. These datasets provide information related to incidents, their causes and severity, emergency response factors, distance, injuries, response strategies, and other operational attributes.

The project involves collecting and understanding the datasets, performing data cleaning and transformation using **Power Query**, and preparing the data for analysis. After preprocessing, **DAX measures and calculated columns** are created to support incident analysis, severity classification, response analysis, and risk assessment.

Interactive Power BI dashboards are then developed to analyze **incident patterns, severity levels, temporal trends, response strategies, overall risk scores, threat exposure, operational vulnerabilities, and resource allocation gaps**. Slicers and interactive visualizations are used to allow users to explore the data from different perspectives.

The overall framework aims to transform raw incident and emergency response data into meaningful visual insights that can support **risk assessment, operational planning, emergency response analysis, and data-driven decision-making**.

Project Approach

The project was carried out using a structured **eight-step approach** to ensure systematic data preparation, analysis, dashboard development, and final reporting.

Step 1: Problem Understanding

The project requirements, problem statement, objectives, scope, and expected outcomes were studied. The main focus was to develop an analytical framework capable of analyzing incident characteristics, severity, response strategies, and risk-related factors through interactive dashboards.

Step 2: Data Collection

Two datasets were selected for the project:

- **Indian Roads Dataset**
- **Integrated Emergency Response Dataset (IERAD)**

The datasets were examined to understand their relevance to incident and emergency response analysis.

Step 3: Data Understanding

The structure and attributes of the datasets were studied to understand the available information and identify the fields required for analysis. Important factors such as **incident causes, severity, time, injuries, distance, response strategy, and incident identifiers** were examined.

Step 4: Data Connection

The datasets were imported into **Microsoft Power BI Desktop** using the data import functionality. The data was then loaded into **Power Query Editor** for further preparation and transformation.

Step 5: Data Preprocessing

Power Query was used to prepare the datasets for analysis. Data types were corrected, required columns were transformed, unnecessary information was removed where required, and analytical fields were prepared. These preprocessing activities improved the consistency and usability of the datasets.

Step 6: Data Modelling

After preprocessing, the datasets were modeled in Power BI. **DAX measures and calculated columns** were created to support the required analysis. Metrics related to incident count, severity, risk, response strategy, injuries, and other analytical requirements were developed.

Step 7: Data Visualization

Interactive Power BI dashboards were developed using **KPI cards, column charts, trend charts, tables, slicers, and other suitable visualizations**. The dashboards were designed to analyze temporal

incident patterns, response strategies, risk levels, threat exposure, operational vulnerabilities, and resource allocation.

Step 8: Analysis, Validation and Reporting

The completed dashboards were analyzed to identify important patterns and insights. Visualizations, calculations, filters, and dashboard interactions were reviewed for accuracy. The final insights, predictive findings, strategic recommendations, documentation, and presentation were then prepared.

Project Flow

Problem Understanding → Data Collection → Data Understanding → Data Connection → Data Preprocessing → Data Modelling → Data Visualization → Analysis & Reporting

Technology Used

Technology / Tool	Purpose
Microsoft Power BI Desktop	Used to develop interactive dashboards and visualize incident, response, and risk-related data.
Power Query Editor	Used for data cleaning, transformation, filtering, and preparation.
Used for data cleaning, transformation, filtering, and preparation.	Used to create calculated columns, measures, KPIs, and risk-related analytical metrics.
Microsoft Excel	Used for reviewing, organizing, and verifying dataset information.
CSV Datasets	Used as the primary data format for the Indian Roads Dataset and Integrated Emergency Response Dataset (IERAD).

Insights from the Dashboard

The developed Power BI dashboards provide meaningful insights into incident patterns, severity, emergency response strategies, and operational risk. The major insights obtained from the analysis are:

- **Incident severity analysis** helps identify high-impact incidents that require greater attention and priority.
- **Temporal analysis** highlights variations in incident activity across different months and time periods, helping identify periods of increased incident occurrence.
- **Cause analysis** helps identify major contributing factors associated with incidents, supporting targeted preventive measures.
- **Response strategy analysis** provides insights into the use of different response strategies based on incident characteristics, distance, and number of injuries.
- **Risk analysis** helps identify incidents with higher overall risk scores and supports prioritization of high-risk situations.
- **Threat exposure analysis** provides an overview of the level of exposure to potential operational threats.
- **Operational vulnerability analysis** helps identify areas where weaknesses may affect effective incident management.
- **Resource allocation analysis** highlights potential gaps between incident requirements and available response resources.
- **Interactive slicers and filters** allow users to explore the dashboards dynamically based on risk levels, response strategies, and other relevant factors.

REAL-WORLD IMPACT FOR MEDIA AND PUBLIC COMMUNICATION

The project can support media and public communication by converting complex incident, risk, and emergency response data into clear and meaningful visual information. The insights generated from the dashboards can help in identifying significant incident trends,

high-risk situations, and response priorities, enabling timely and accurate communication. It can also improve public awareness by presenting important risk patterns and safety concerns in an understandable manner. The visual dashboards can help media organizations communicate incident-related information more effectively to the public. They can also support authorities in sharing reliable information during critical situations and improving transparency. Overall, the framework can contribute to evidence-based reporting, timely emergency communication, public awareness, and better coordination between authorities, emergency response teams, media, and the public.

MILESTONE 1 – THREAT CLASSIFICATION & SEVERITY ANALYTICS

Objective

The objective of Milestone 1 was to develop an interactive **Threat Classification and Severity Analytics Dashboard** using Power BI. The milestone focused on cleaning and standardizing the datasets, modeling the data, classifying incidents, and analyzing incident severity, response performance, impact, and geographical distribution.

Milestone Explanation

During Milestone 1, the datasets were prepared and analyzed using **Microsoft Power BI** to develop an interactive Threat Classification and Severity Analytics Dashboard. The data was cleaned, standardized, and modeled to support meaningful analysis of incident characteristics and response-related metrics. The dashboard was designed with KPI cards and interactive visualizations to provide an overall view of the incident data.

The dashboard includes KPI cards for **Total Incidents, Critical Incidents, Threat Severity Index, Average Response Time, Total Cities, Total Escalated Incidents, and Average Risk Score**. These KPIs provide a quick summary of the overall incident and response situation.

Several visualizations were developed to analyze the data from different perspectives. **Incident Count by Category** compares incidents across causes such as distraction, speeding, weather, drunk driving, and poor road conditions. **Monthly Incident Trend** shows the variation in incident activity over time, while **Incident Severity Distribution** presents the proportion of minor, major, and fatal incidents. **Total Incidents by Day of Week** helps compare incident occurrence across different days.

The dashboard also includes **Average Response Time by Incident Cause** to compare response performance across categories and **Incident Impact Distribution** to understand the distribution of low, high, and critical impacts. **Incident Count by State** and **Count of Escalation by State** provide geographical comparisons of incidents and escalated cases.

Interactive slicers for **Incident Date, Incident Cause, Incident Severity, City, and State** allow users to filter the dashboard and explore specific subsets of the data. These interactive features support detailed analysis and help decision-makers identify important incident patterns, severity levels, response performance, and geographical differences. The milestone requirements specifically called for KPI cards, analytical charts, and interactive slicers for these types of analysis.

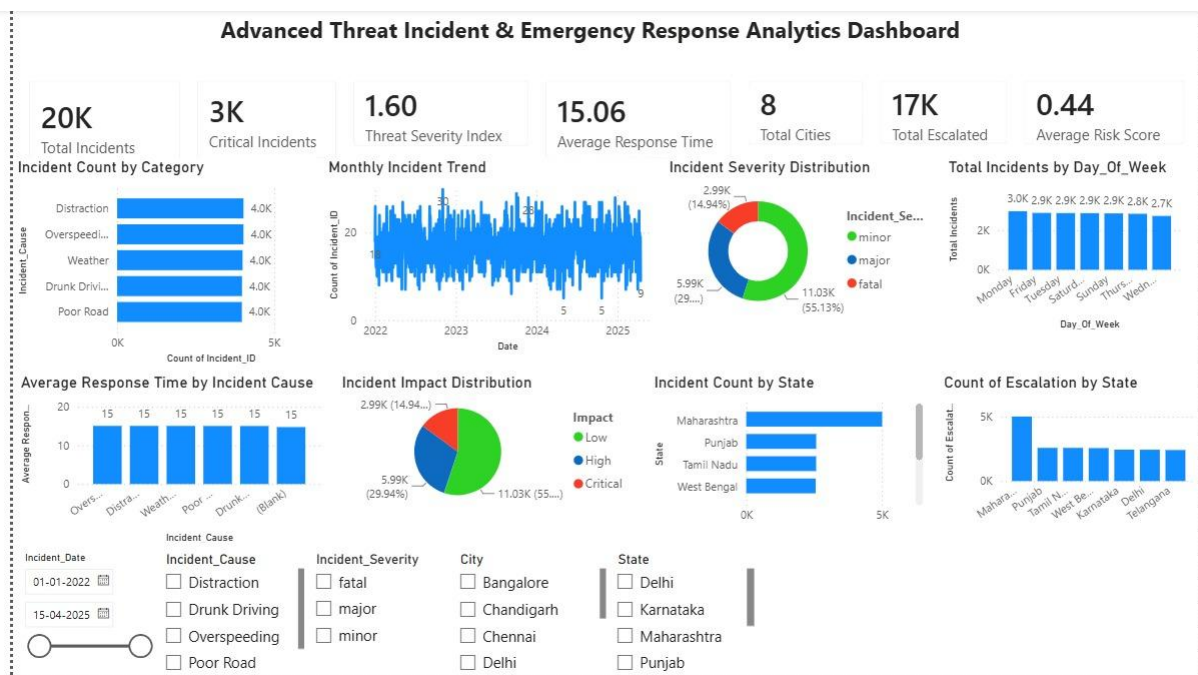
Outcome

The Milestone 1 dashboard provides an interactive analytical view of incident classification, severity, response performance, impact, and geographical distribution. It establishes the foundation for the subsequent milestones by converting the prepared incident data into meaningful visual insights for risk monitoring and decision-making.

Key Visualizations Used

- **KPI Cards:** Overall incident and risk summary
- **Bar Chart:** Incident Count by Category
- **Line Chart:** Monthly Incident Trend
- **Donut Chart:** Incident Severity Distribution

- **Column Chart:** Total Incidents by Day of Week
- **Column Chart:** Average Response Time by Incident Cause
- **Donut Chart:** Incident Impact Distribution
- **Bar Chart:** Incident Count by State
- **Column Chart:** Count of Escalation by State
- **Slicers:** Date, Incident Cause, Incident Severity, City, and State



MILESTONE 2 – THREAT CLASSIFICATION & GEOSPATIAL INTELLIGENCE

Objective

The objective of Milestone 2 was to extend the Milestone 1 dashboard by incorporating **Threat Classification and Geospatial Threat Intelligence** capabilities. The dashboard was developed to help analyze incident severity, geographical distribution, response performance, threat concentration, and high-risk locations through interactive Power BI visualizations.

Milestone Explanation

In Milestone 2, the existing Power BI dashboard developed in Milestone 1 was enhanced with **threat classification and geospatial intelligence features**. The focus was on understanding the severity,

geographical distribution, concentration, and operational impact of incidents across different cities and states.

The first dashboard page provides an overview of incident and severity analysis. KPI cards display **Total Incidents, Critical Incidents, Average Response Time, and Total Cities**. The **Severity Distribution** visual shows the distribution of fatal, major, and minor incidents, while **Incident Category Analysis** compares incidents across different causes such as distraction, speeding, weather, and drunk driving.

The dashboard also includes **Average Response Time by City** to compare response performance across locations. **Top 8 Cities by Incidents** highlights cities with higher incident counts, while the **City-wise Incident & Response Summary** provides a detailed comparison of total incidents and average response time for each city.

Geospatial analysis was incorporated through **Risk Surface Analysis** and **City-wise Incident Hotspots**, which provide a geographical view of incident concentration. These visuals help identify areas with greater incident activity and support location-based analysis. The milestone requirements specifically included bubble-map and filled-map based geographical analysis, top risk cities, response time by city, and threat concentration analysis.

The second dashboard page focuses on deeper **hotspot and threat concentration analysis**. **Hotspot Evolution Analysis** compares incident activity across cities and months to observe how hotspot patterns change over time. **Threat Concentration by City** presents the distribution of incident severity across different cities, while **Environmental Risk Indicators** compares incident occurrence under environmental conditions such as clear weather, rain, and fog.

The **Spatial Clustering & Hotspot Evolution Analysis** categorizes cities based on their incident concentration, while the **Threat Concentration Zone** visualization compares the total incidents across the identified cities. These visualizations provide additional support for identifying locations with comparatively higher incident concentrations.

Interactive reporting features were also incorporated through **City, Incident Severity, and Incident Cause slicers**, allowing users to filter and explore the dashboard dynamically. The official milestone requirements included city, severity, and category slicers along with interactive visual features such as drill-through, tooltips, dynamic titles, and visual interactions.

Outcome

Milestone 2 enhanced the basic incident dashboard into a more comprehensive **Threat Classification and Geospatial Intelligence Dashboard**. It provided a detailed view of incident severity, category distribution, city-wise response performance, incident hotspots, environmental indicators, spatial concentration, and changing hotspot patterns, supporting more effective geographical and operational analysis.

Key Visualizations Used

- **KPI Cards:** Total Incidents, Critical Incidents, Average Response Time, Total Cities
- **Donut Chart:** Severity Distribution
- **Bar Chart:** Incident Category Analysis
- **Column Chart:** Average Response Time by City
- **Bar Chart:** Top Cities by Incidents
- **Map Visuals:** Risk Surface Analysis and City-wise Incident Hotspots
- **Matrix:** City-wise Incident & Response Summary
- **Line Chart:** Hotspot Evolution Analysis
- **Stacked Column Chart:** Threat Concentration by City
- **Bar Chart:** Environmental Risk Indicators
- **Bar Chart:** Spatial Clustering & Hotspot Evolution
- **Bar Chart:** Threat Concentration Zone
- **Slicers:** City, Incident Severity, and Incident Cause

Threat Classification & Geospatial Intelligence Dashboard - All Cities

Dashboard Title

20K

Total Incidents

3K

Critical Incidents

15.06

Average Response Time

8

Total Cities

City

- ☐ Bangalore
☐ Chandigarh
☐ Chennai

Incident_Severity

- ☐ fatal
☐ major
☐ minor

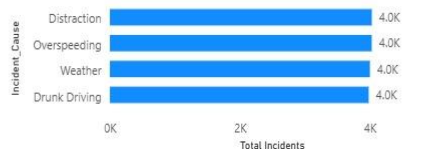
Incident_Cause

- ☐ Distraction
☐ Drunk Driving
☐ Overspeeding

Severity Distribution



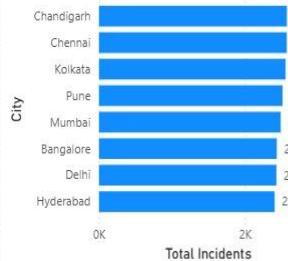
Incident Category Analysis



Average Response Time by City



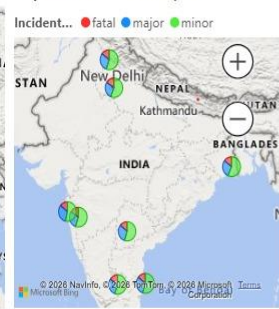
Top 8 Cities by Incidents



Risk Surface Analysis



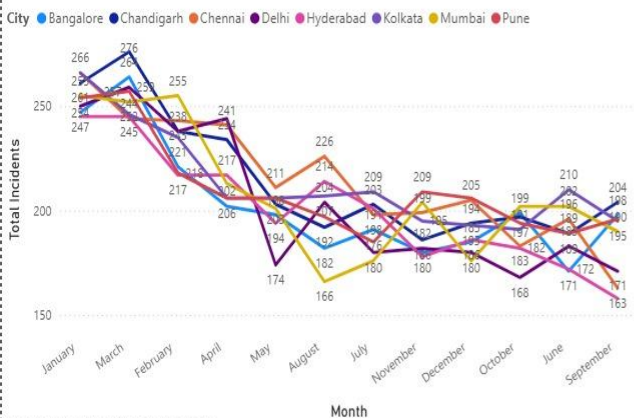
City Wise Incident Hotspots



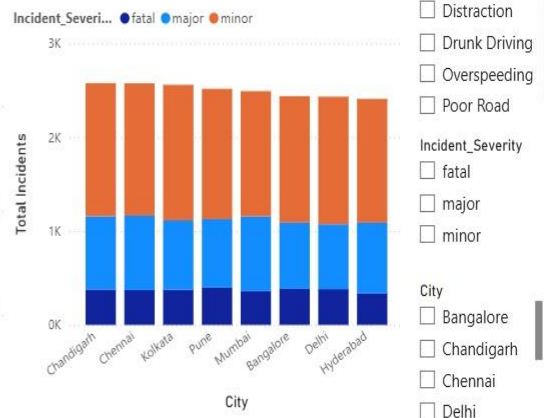
City-wise Incident & Response Summary

City	Average Response Time	Total Incidents
Chandigarh	15.05	2577
Chennai	15.08	2575
Kolkata	15.05	2559
Pune	15.09	2517
Mumbai	15.06	2492
Bangalore	15.02	2438
Delhi	15.07	2433
Hyderabad	15.04	2409
	14.76	
Total	15.06	20000

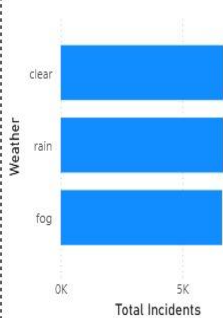
Hotspot Evolution Analysis



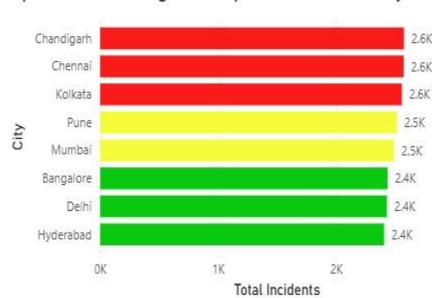
Threat Concentration by City



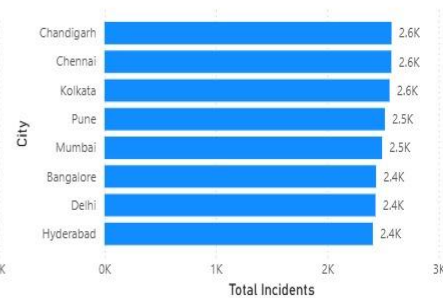
Environmental Risk Indicators



Spatial Clustering & Hotspot Evolution Analysis



Threat Concentration Zone



MILESTONE 3 – TEMPORAL INCIDENT ANALYTICS & INCIDENT RESPONSE ANALYTICS

Objective

The objective of Milestone 3 was to extend the Milestone 2 Power BI solution by implementing **Temporal Incident Analytics** and **Incident Response Analytics**. The milestone focused on identifying peak incident periods, emerging threat patterns, environmental factors, operational bottlenecks, and the effectiveness of incident response.

Milestone Explanation

In Milestone 3, the existing Power BI project from Milestone 2 was enhanced to analyze **when incidents occur and how effectively they are handled**. Two interactive dashboard pages were developed: **Temporal Incident Analytics** and **Incident Response Analytics**.

The **Temporal Incident Analytics** dashboard provides a detailed analysis of incident occurrence across different time dimensions. The **Emerging Threat Pattern Analysis** visual compares incident causes across months and helps identify changes in incident patterns. **Hourly Incident Trend** analyzes incident activity across different hours of the day, while **Daily Incident Trend** compares incident counts across days of the week. **Weekly Incident Trend** provides a week-wise view of incident activity, and **Monthly Incident Trend** compares incident occurrence across months. These visualizations support the identification of periods with higher or changing incident activity. The milestone requirements specifically included hourly, daily, weekly, monthly, seasonal, anomaly/activity-spike, environmental-factor, and emerging-threat analysis.

The dashboard also includes **Seasonal Trend Analysis**, which categorizes incidents according to seasons such as monsoon, winter, summer, and post-monsoon. The **Environmental Factor Analysis**

compares incident severity across different shifts, including morning, evening, and night. **Activity Spike Analysis** highlights unusual variations in incident activity and helps identify potential activity spikes.

The **Incident Response Analytics** dashboard focuses on evaluating the operational handling of incidents. KPI cards display **Total Responses, Average Response Time, and Average Resolution Time**. **Incident Status Distribution** compares open and closed incidents, while **Response Time by Incident Severity** compares response times for fatal, major, and minor incidents.

The dashboard further analyzes response operations through **Incidents Handled by Response Team, Average Response Time Across Teams, and Average Resolution Time Across Teams**. **Average Response Time Across Cities** provides a city-level comparison of response performance, while **Response Strategy Distribution** shows the distribution of strategies such as **Ambulance Only, Drone Only, and Hybrid Dispatch**. These analyses help evaluate response performance and identify potential operational differences. The milestone requirements included response volume, response and resolution time, open/closed status, response time by team, city and severity, and response status analysis.

Interactive slicers were incorporated to allow users to filter the dashboards based on **Year, Month, Incident Severity, Incident Category, City, and Environmental Factors**. These filters enable users to examine specific periods, locations, categories, and severity levels and support interactive exploration of the dashboard.

Outcome

Milestone 3 resulted in two interactive analytical dashboards that provide a combined view of **incident timing and response performance**. The Temporal Incident Analytics dashboard helps identify incident trends, activity spikes, seasonal patterns, and environmental influences, while the Incident Response Analytics

dashboard helps evaluate response time, resolution time, incident status, response teams, cities, severity levels, and response strategies.

Key Visualizations Used

Temporal Incident Analytics

- Emerging Threat Pattern Analysis
- Hourly Incident Trend
- Daily Incident Trend
- Weekly Incident Trend
- Monthly Incident Trend
- Seasonal Trend Analysis
- Activity Spike Analysis
- Environmental Factor Analysis

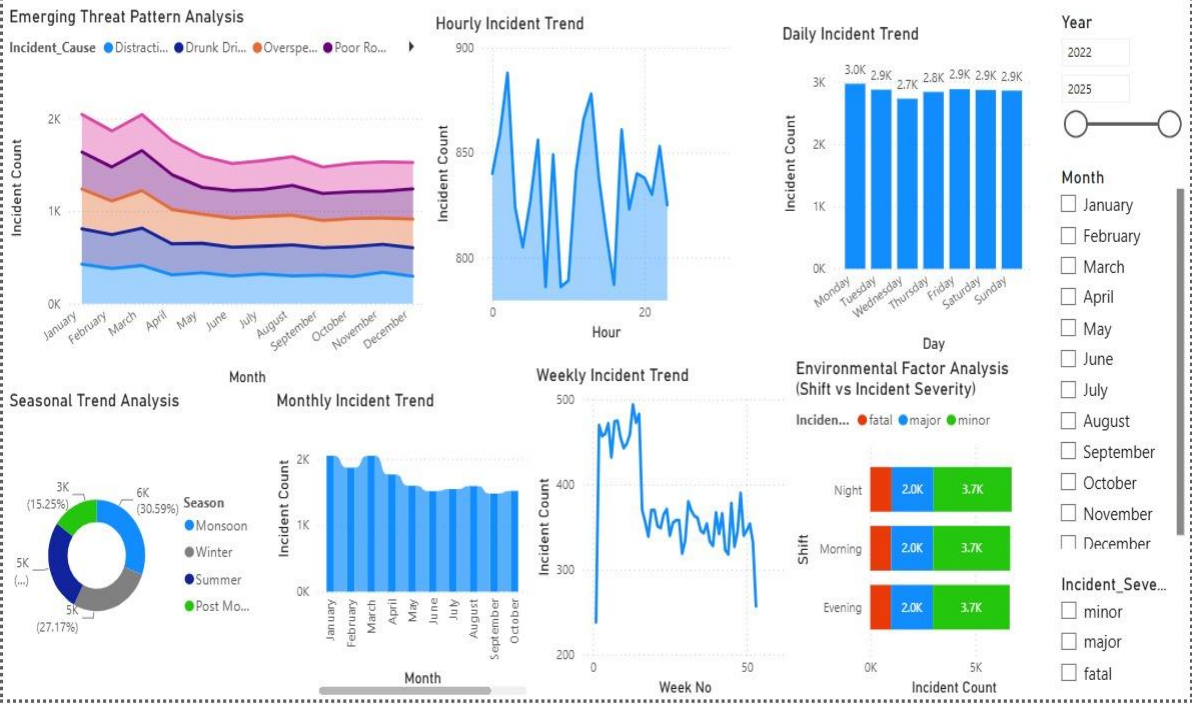
Incident Response Analytics

- KPI Cards
- Incident Status Distribution
- Response Time by Incident Severity
- Incidents Handled by Response Team
- Average Response Time Across Teams
- Average Resolution Time Across Teams
- Average Response Time Across Cities
- Response Strategy Distribution

Interactive Filters

- Year
- Month
- Incident Severity
- Incident Category
- City
- Environmental Factor

Temporal Incident Analytics



Incident Response Analytics



MILESTONE 4 – SECURITY RISK ASSESSMENT, PREDICTIVE INTELLIGENCE & EXECUTIVE ANALYTICS

Objective

The objective of Milestone 4 was to extend the Milestone 3 Power BI solution beyond historical incident and response analysis by implementing **Security Risk Assessment, Predictive Intelligence, and Executive Analytics**. The milestone focused on measuring threat exposure, identifying operational vulnerabilities and resource requirements, comparing risks across locations and response strategies, forecasting emerging risks, and providing executive-level decision support.

Milestone Explanation

In Milestone 4, the existing Power BI project was enhanced with advanced **risk assessment and predictive analytics capabilities**. Three dashboards were developed: **Security Risk Assessment, Predictive Intelligence, and Executive Intelligence**. These dashboards transformed the existing analytical solution into a more comprehensive framework for assessing current security risks and supporting future-oriented decision-making.

The **Security Risk Assessment** dashboard provides an overall view of the security risk situation using KPI cards for **Overall Risk Score, Threat Exposure, Critical Incidents, High-Risk Incidents, and Resource Demand**. The **Risk by Operational Unit** visualization compares risk levels across different response strategies, while **Threat Exposure Distribution** shows the distribution of moderate, high, and critical risk levels.

The dashboard also includes **Top 7 High-Risk Locations**, which identifies cities with comparatively higher overall risk scores. **Operational Vulnerability Analysis** provides a comparison of incidents across different risk levels, while **Security Weakness Analysis** highlights major incident causes associated with security weaknesses. **Resource Allocation Demand** compares resource requirements across response strategies. These analyses support the

identification of risk concentration, vulnerabilities, and potential resource gaps, which are key requirements of the milestone.

The **Predictive Intelligence** dashboard focuses on future-oriented risk analysis. KPI cards display **High-Risk Predicted Incidents**, **Prediction–Severity Agreement**, **Critical Risk Percentage**, and **Operational Vulnerability Score**. The **Predicted vs Actual Severity** matrix compares predicted risk levels with actual incident severity, providing an evaluation of prediction performance.

The **Emerging Risk Score Forecast** visual presents the variation and forecast of risk scores over time, while **Incident Volume Forecast** provides a forecast of future incident volumes. **Predicted Threat Risk Distribution** shows the distribution of incidents across predicted risk levels. The dashboard also identifies **Top 5 High-Risk Locations** and presents **Average Predicted Risk Probability**, supporting the identification of potential future high-risk areas. These analyses correspond to the milestone requirements for predictive threat risk modeling, incident risk prediction, emerging-risk forecasting, incident trend forecasting, high-risk zone identification, and model performance evaluation.

The **Executive Intelligence Dashboard** summarizes the most important risk and operational indicators for high-level decision-making. KPI cards present the **Overall Risk Score**, **Critical Incidents**, **Resource Demand**, **Critical Risk Percentage**, and **Operational Vulnerability Score**. The **High-Risk Location** visualization identifies locations with higher risk scores, while **Emerging Risk Zones** summarizes the distribution of moderate, high, and critical risks.

The **Strategic Risk Overview** provides a city-wise comparison of incident count, average risk score, and resource demand. **Resource Demand by Response Strategy** compares the resource requirements of Drone Only, Ambulance Only, and Hybrid Dispatch strategies. **Predicted Risk Locations** provides a comparative view of predicted risk probabilities across cities. Together, these visuals provide executive-level metrics for understanding current risks, future risk

patterns, resource requirements, and strategic priorities. The executive dashboard requirements specifically included overall risk score, high-risk locations, critical risk indicators, predicted risk, emerging risk zones, operational indicators, and strategic decision-support metrics.

Interactive filters were incorporated across the dashboards to support dynamic analysis. The milestone requirements included **Year, Location/City, Severity, Category, Operational Unit, and Risk Level slicers**, with the requirement that dashboard visuals interact correctly.

Outcome

Milestone 4 enhanced the project from historical incident and response analysis into an integrated **security risk and predictive intelligence framework**. The developed dashboards provide insights into current risk levels, threat exposure, operational vulnerabilities, security weaknesses, resource demand, predicted risks, emerging risk zones, and high-risk locations. The Executive Intelligence Dashboard further consolidates these findings into strategic metrics that can support **risk prioritization, resource planning, operational decision-making, and future-oriented security management**.

Key Visualizations Used

Security Risk Assessment

- KPI Cards
- Risk by Operational Unit
- Threat Exposure Distribution
- Top High-Risk Locations
- Security Weakness Analysis
- Operational Vulnerability Analysis
- Resource Allocation Demand

Predictive Intelligence

- KPI Cards
- Predicted vs Actual Severity

- Emerging Risk Score Forecast
- Incident Volume Forecast
- Predicted Threat Risk Distribution
- Top High-Risk Locations
- Average Predicted Risk Probability

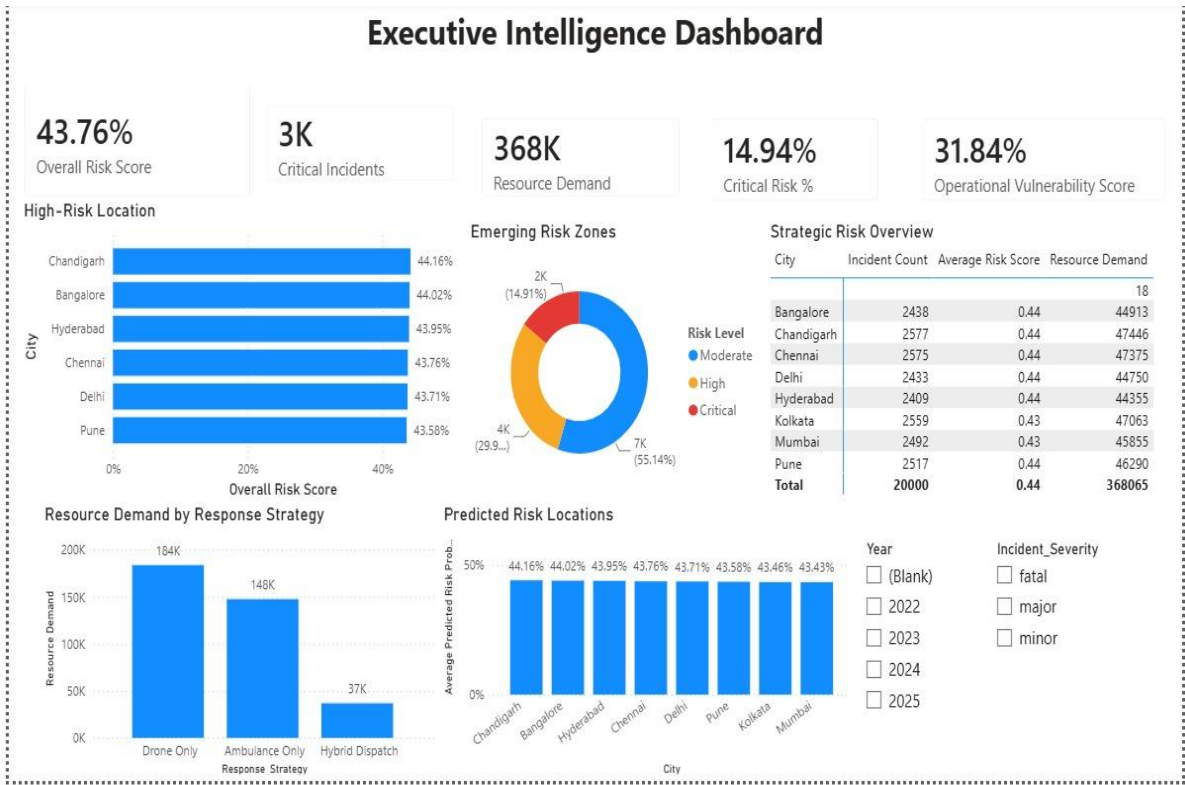
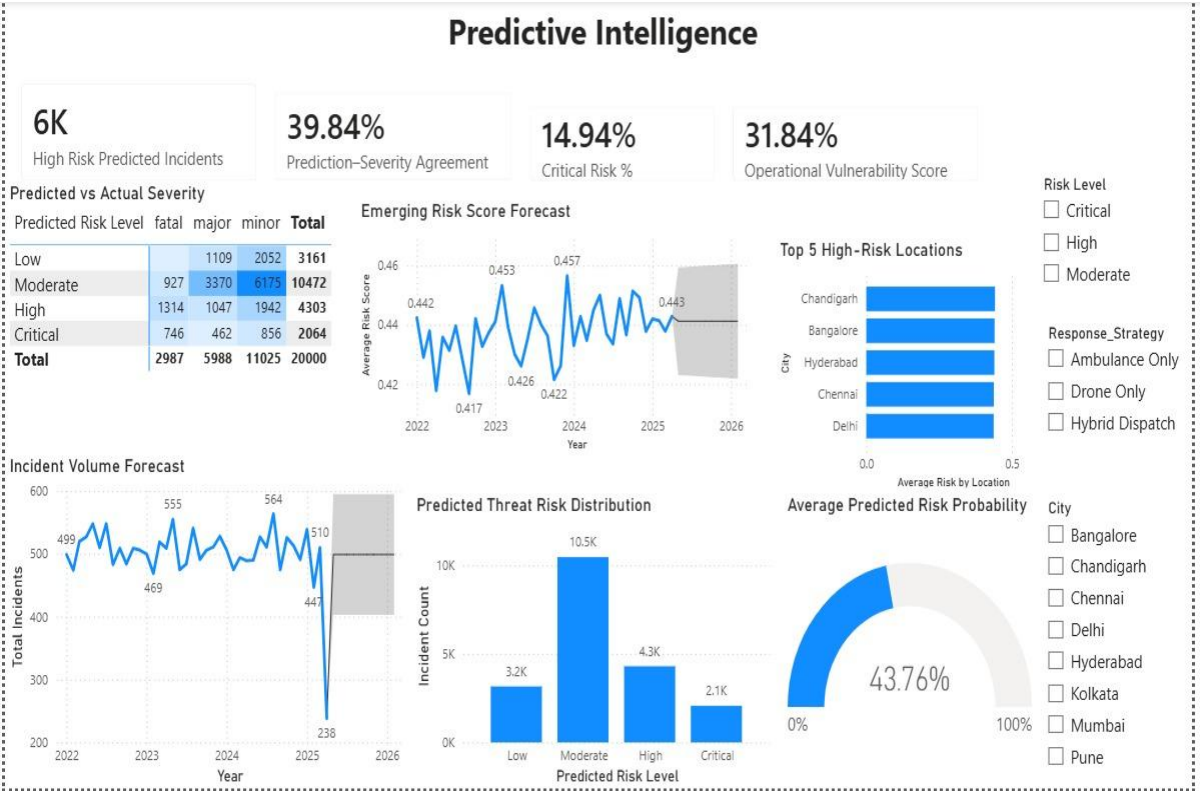
Executive Intelligence

- KPI Cards
- High-Risk Location
- Emerging Risk Zones
- Strategic Risk Overview
- Resource Demand by Response Strategy
- Predicted Risk Locations

Interactive Filters

- Year
- City
- Incident Severity
- Incident Cause
- Response Strategy / Operational Unit
- Risk Level





PROJECT EXECUTION DETAILS – STEP-BY-STEP PROCESS

The project was executed through the following systematic steps:

Step 1: Problem Understanding

The project requirements and objectives were studied to understand the need for analyzing incident patterns, emergency response performance, geographical risks, security risks, and predictive indicators.

Step 2: Data Collection

Two datasets were selected for the project: **Indian Roads Dataset** and **Integrated Emergency Response Dataset (IERAD)**.

Step 3: Data Understanding

The datasets were examined to understand their structure, attributes, data types, and relevant fields required for incident, severity, response, and risk analysis.

Step 4: Data Import

The datasets were imported into **Microsoft Power BI Desktop** for further processing, modeling, and visualization.

Step 5: Data Cleaning and Transformation

The data was cleaned and transformed using **Power Query**. Data types were standardized, required fields were prepared, and the datasets were made suitable for analysis.

Step 6: Data Modelling

The prepared data was modeled in Power BI to support analysis across incident, response, geographical, temporal, and risk-related dimensions.

Step 7: DAX Calculations

DAX measures and calculated columns were created to calculate important analytical metrics such as incident counts, severity indicators, response measures, risk scores, threat exposure, vulnerabilities, and resource demand.

Step 8: Dashboard Development

Interactive Power BI dashboards were developed to analyze **threat classification, severity, geospatial patterns, temporal trends, incident response, security risks, and predictive intelligence.**

Step 9: Data Visualization

Suitable visualizations such as **KPI cards, bar charts, column charts, line charts, donut charts, maps, tables, and slicers** were used to present the analytical results clearly.

Step 10: Interactive Analysis

Slicers and filters were added to enable users to analyze the dashboards based on parameters such as **year, city, incident cause, incident severity, response strategy, and risk level.**

Step 11: Insight Generation

The completed dashboards were analyzed to identify important **incident patterns, high-risk locations, response trends, operational vulnerabilities, resource requirements, and emerging risks.**

Step 12: Final Reporting

The findings were compiled into the final **documentation and presentation**, including key insights, predictive findings, strategic recommendations, and overall project outcomes.

CHALLENGES FACED

During the development of the project, several challenges were encountered while working with the datasets, performing analysis, and developing the Power BI dashboards.

- **Data Preparation:** Cleaning and transforming the raw datasets required careful handling of data types, missing values, and inconsistent fields.
- **Data Modelling:** Establishing an appropriate data model and working with the incident and emergency response datasets required careful consideration of common fields and relationships.
- **DAX Calculations:** Creating calculated columns and measures for severity, risk scores, threat exposure, operational vulnerability, and resource demand required careful formulation and validation.
- **Visualization Selection:** Selecting suitable visualizations for different analytical requirements while keeping the dashboards clear and easy to understand was challenging.
- **Dashboard Design:** Organizing multiple KPIs, charts, maps, slicers, and analytical components within limited dashboard space required several design adjustments.
- **Interactive Filtering:** Ensuring that slicers and filters worked correctly across different dashboard visuals required testing and refinement.
- **Predictive Analysis:** Developing and presenting predictive risk, incident volume, and emerging-risk analysis required careful interpretation of the available data and analytical results.
- **Insight Generation:** Converting the dashboard outputs into meaningful operational insights and strategic recommendations required detailed analysis of the visualized results.

LEARNINGS & SKILLS ACQUIRED

Technical Skills

- Gained practical experience in **Microsoft Power BI** for developing interactive dashboards.
- Learned **Power Query** for data cleaning, transformation, and preparation.
- Developed skills in **DAX** for creating calculated columns, measures, and analytical metrics.
- Learned **data modelling** and establishing relationships between datasets.
- Improved knowledge of **data visualization**, including KPI cards, charts, maps, tables, and slicers.
- Gained experience in developing interactive dashboards with filters and visual interactions.

Analytical & Problem-Solving Skills

- Developed the ability to analyze **incident patterns, severity levels, and temporal trends**.
- Learned to identify **high-risk incidents, geographical hotspots, and emerging risk patterns**.
- Gained experience in analyzing **response performance, operational vulnerabilities, and resource allocation**.
- Improved the ability to interpret complex datasets and convert them into meaningful insights.
- Developed problem-solving skills by identifying data and dashboard issues and implementing suitable solutions.

Future Enhancements

- Integration of **real-time incident and emergency response data**.
- Implementation of advanced **machine learning-based predictive models**.
- Development of automated alerts for high-risk incidents.

- Enhancement of forecasting for emerging risks and incident trends.
- Integration of additional environmental and contextual data sources.
- Development of advanced executive dashboards for real-time situational awareness.

Soft Skills

- Improved **communication and presentation skills** through dashboard demonstrations and project presentations.
- Developed **teamwork and collaboration** skills while working on the internship project.
- Improved **time management** through milestone-based project execution.
- Developed better **documentation and reporting** skills.
- Gained confidence in explaining analytical findings and recommendations.

Domain Knowledge and Application

- Gained knowledge of **incident and threat classification** and severity assessment.
- Understood the importance of **emergency response analysis** and operational performance.
- Learned how data can be used for **security risk assessment and resource planning**.
- Developed an understanding of **threat exposure, operational vulnerabilities, and emerging risks**.
- Learned how business intelligence and data visualization can support **risk management, emergency planning, and data-driven decision-making**.

CONCLUSION

The **Cyber Threat Analytics Platform with Incident Visualization System** successfully transformed incident and emergency response data into meaningful and interactive analytical dashboards using **Microsoft Power BI**. The project involved data collection, preprocessing, transformation, modelling, DAX-based analysis, visualization, and interpretation of results using the **Indian Roads Dataset** and **Integrated Emergency Response Dataset (IERAD)**.

The developed solution provides comprehensive analysis of **incident severity, geographical patterns, temporal trends, response performance, security risks, operational vulnerabilities, resource requirements, and predictive risk indicators**. The dashboards enable users to explore the data interactively and identify important patterns, high-risk locations, emerging risks, and operational areas requiring attention.

Through the project, practical skills in **Power BI, Power Query, DAX, data visualization, data modelling, analytical problem-solving, and dashboard development** were gained. The project demonstrates how data analytics and business intelligence can be applied to support **risk assessment, emergency response planning, resource allocation, and data-driven decision-making**.

Overall, the project provides a structured analytical framework that moves from **historical incident analysis to risk assessment and predictive intelligence**, creating a strong foundation for future enhancements such as real-time data integration, advanced machine learning models, automated alerts, and improved predictive risk analysis.